# CISplatin (100mg/m$^2$) with Radiotherapy (RT)-21 day

## INDICATIONS FOR USE:

| INDICATION | ICD10 | Regimen Code | HSE approved Reimbursement Status* |
|---|---|---|---|
| Chemoradiation treatment for locally advanced (stage III to IV) squamous cell carcinoma (SCC) of the head and neck | C76 | 00387a | N/A |

*This applies to post 2012 indications

## TREATMENT:

*The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.*

CISplatin is administered once every 21 days with concurrent radiotherapy for 3 cycles. Treatment should continue for a total of 3 cycles unless the patient experiences disease progression or unacceptable toxicity.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

| Day | Drug | Dose | Route | Diluent & Rate | Cycle |
|---|---|---|---|---|---|
| 1 | CISplatin | 100mg/m$^2$ | IV Infusion | 1000mL NaCl 0.9% over 2 hours (Pre and Post hydration therapy required)** | Every 21 days |
| **Pre and post hydration therapy required for CISplatin<br>See local hospital policy recommendations.<br>Suggested prehydration for CISplatin therapy:<br>Administer 10mmol magnesium sulphate ($MgSO_4$) ((+/-KCl 10-20mmol/L if indicated) in 1000mL NaCl 0.9% over 60-120 minutes (Refer to relevant local hospital policy for advice on administration of electrolyte infusions) (3).<br><br>Administer CISplatin as described above<br>Post hydration: Administer 1000mL 0.9% NaCl over 60 minutes | | | | | |
| Mannitol 10% may be used as per local policy to induce diuresis, although there is no conclusive evidence that this is required. The routine use of furosemide to increase urine flow is not recommended unless there is evidence of fluid overload (4, 5). | | | | | |
| **CISplatin (radiosensitizer) – Radiotherapy**<br>Since CISplatin is used in this regimen as a radiosensitising agent, it is to be administered on a day on which radiotherapy is delivered. Radiotherapy should start after CISplatin infusion is completed. If radiotherapy is cancelled on the CISplatin day, do not give CISplatin that day and postpone chemotherapy until radiation therapy resumes. | | | | | |

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

## ELIGIBILITY:

- Indications as above
- ECOG 0-2
- Adequate hepatic, renal, and bone marrow function

## EXCLUSIONS:

- Hypersensitivity to CISplatin or any of the excipients
- Significant hearing impairment/tinnitus
- Pregnancy
- Breastfeeding



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## CAUTIONS:

- Pre existing renal impairment – see Table 2

## PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist

## TESTS:

**Baseline tests**:

- FBC, liver and renal profiles
- Audiology referral and creatinine clearance if clinically indicated.

**Regular tests**:

- FBC, liver and renal profiles prior to each cycle.

**Disease monitoring:**

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

## DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant.

**Haematological:**

**Table 1: Dose modification of CISplatin in haematological toxicity**

| ANC ($x10^9$/L) | | Platelets ($x10^9$/L) | Dose |
|---|---|---|---|
| ≥1.5 | **and** | ≥ 100 | 100% |
| 1 -1.49 | **or** | 75-99 | 75% |
| <1 | **or** | <75 | Delay chemoradiation until recovery |

**Renal and Hepatic Impairment:**

**Table 2: Dose modification of CISplatin in renal and hepatic impairment**

| **Renal Impairment** | | **Hepatic Impairment** |
|---|---|---|
| **CrCl (mL/min)** | **Dose** | No need for dose adjustment is expected |
| 50-59 | 75% of original dose | |
| 40-49 | 50% of original dose | |
| <40 | Not recommended | |
| Haemodialysis | 50% of the original dose may be considered | |
| Renal and hepatic recommendations as per Giraud et al 2023 | | |



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**Management of adverse events:**

**Table 3: Dose Modification of CISplatin for Adverse Events**

| Adverse reactions | Recommended dose modification |
|---|---|
| Peripheral neuropathy<br>Grade 2 | Reduce CISplatin dose by 25% |
| Grade 3 or 4 | Omit CISplatin |

## SUPPORTIVE CARE:

### EMETOGENIC POTENTIAL

- As outlined in NCCP Classification Document for Systemic Anti Cancer Therapy (SACT) Induced Nausea and Vomiting- Available on the NCCP website

**CISplatin:** High (**Refer to local policy).**

**For information:**

**Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:**

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

### PREMEDICATIONS:

### OTHER SUPPORTIVE CARE:

- Hydration pre and post CISplatin administration (Refer to local policy or see recommendations above).
- Oral hydration should be encouraged for 24 hours.
- Prior to initiation of treatment, patients should be referred for consultation to Dentistry and Nutrition Services.

## ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details
- Neutropenia: Fever or other evidence of infection must be assessed promptly and treated appropriately.

## REGIMEN SPECIFIC COMPLICATIONS:

- Ototoxicity and sensory neural damage should be assessed by history prior to each cycle**.**
- Peripheral neuropathy: Peripheral neuropathy has been reported with CISplatin. The patient should be monitored for symptoms of peripheral neuropathy. Patients experiencing new or worsening peripheral neuropathy may require dose modification.



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## DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

## REFERENCES:


1. Forastiere AA, Zhang Q, Weber RS et al. Long-term results of RTOG 91-11: a comparison of three nonsurgical treatment strategies to preserve the larynx in patients with locally advanced larynx cancer. J Clin Oncol. 2013; 31(7):845.
2. Forastiere AA, Goepfert H, Maor M et al. Concurrent chemotherapy and radiotherapy for organ preservation in advanced laryngeal cancer. N Engl J Med. 2003;349(22):2091
3. Perazella MA et al. CISplatin nephrotoxicity. UptoDate.Last updated April 2024. Available at: https://www.uptodate.com/contents/cisplatin-nephrotoxicity?search=portilla%20cisplatin&source=search_result&selectedTitle=4%7E150&usage_type=default&display_rank=4
4. Irish Medication Safety Network: Best Practice Guidelines for the Safe Use of Intravenous Potassium in Irish Hospitals. Published Oct 2020. Available at: https://imsn.ie/wp-content/uploads/2020/10/IMSN-Best-Practice-Guideline-on-IV-Potassium-Oct-2020-approved.pdf
5. Nephrotoxicity Associated with CISplatin EviQ ID: 184 v.4 https://www.eviq.org.au/clinical-resources/side-effect-and-toxicity-management/prophylaxis-andprevention/184-nephrotoxicity-associated-with-CISplatin 4.
6. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
7. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
8. CISplatin 1mg/mL Concentrate for Solution for Infusion. Summary of Product Characteristics. Last updated 14/10/2025.Accessed November 2025. Available at: https://www.hpra.ie/find-a-medicine/for-human-use/authorised-medicines/details/27566


| Version | Date | Amendment | Approved By |
|---|---|---|---|
| 1 | 20/12/2016 | | Prof Maccon Keane |
| 2 | 26/11/2018 | Updated to new NCCP template. Updated with revised hydration regimen for CISplatin | Prof Maccon Keane |
| 3 | 10/11/2020 | Reviewed | Prof Maccon Keane |
| 4 | 02/09/2021 | Updated cisplatin hydration recommendation | Prof Maccon Keane |
| 5 | 05/12/2025 | Regimen reviewed. Updated exclusions and baseline testing sections. Added cautions section. Updated renal and hepatic dose modifications table to align with Giraud et al 2023. Updated other supportive care section.Updated regimen specific complications section. Updated regimen in line with NCCP standardisation. | Dr Colm Mac Eochagain |

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.



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